29 · The male reproductive system: every structure and its job

The words of this track, as it says them. Bold brown is the examiner.

The male reproductive system. Ten marks.

Label all the structures of the male reproductive system, and state their functions.

This is the male half of her biggest written question: ten marks on her student-marked quiz, and her model answer is a link to her own notes. So here are her notes, organ by organ, in the order the sperm travel. She'll give you a side-on diagram, so picture it as you listen.

The hook: make, store, carry, add, deliver. The testes make, the epididymis stores, the ducts carry, three glands add fluid, and the penis delivers.

The system's function: to produce the gametes, the sperm; to secrete the male sex hormones, mainly testosterone; and to deliver the sperm into the female reproductive tract.

The scrotum: a skin sac outside the body cavity, holding the two testes. Hanging outside keeps them about three degrees cooler than the core, which sperm production needs. Two muscles hold that temperature steady. The dartos is smooth muscle that wrinkles the skin, to cut heat loss. The cremaster is skeletal muscle that pulls the testes up towards the body when it's cold.

The testes: the male gonads. Inside, the seminiferous tubules produce the sperm. Between the tubules, the Leydig cells make testosterone. And the Sertoli cells nourish the developing sperm, and make inhibin.

The epididymis: a cup-shaped duct on the back of each testis, with a head, a body and a tail. It stores the sperm, and it's where they mature and become motile. At ejaculation it contracts, and pushes them on into the vas deferens.

The vas deferens, also called the ductus deferens: the duct that carries the sperm from the epididymis up into the pelvis, where it widens into the ampulla. A vasectomy cuts the vas deferens, so sperm can't get from the testis to the outside, though fluid from the seminal vesicles and the prostate still leaves.

The seminal vesicles: paired glands on the back wall of the bladder. Each one joins its vas deferens to form an ejaculatory duct. They make about sixty per cent of the semen: a sticky, alkaline fluid, with fructose to feed the sperm, and prostaglandins.

The prostate: a doughnut-shaped gland under the bladder, wrapped around the urethra. It adds about a third of the semen: a milky fluid with citric acid as an energy source, enzymes, and prostate-specific antigen, PSA.

The bulbourethral glands, also called Cowper's glands: pea-sized, just below the prostate. Before ejaculation they release a thick, clear mucus that neutralises any traces of acidic

urine in the urethra.

The urethra: it carries both urine and semen, at different times, through the one opening. It has three parts: prostatic, through the prostate; membranous; and spongy, or penile, through the penis.

The penis: it delivers the sperm into the female reproductive tract. It has a root, a shaft, and the glans at the end, covered by the prepuce, the foreskin. Inside are three columns of erectile tissue: the two corpora cavernosa on top, and the corpus spongiosum, which surrounds the urethra and forms the glans.

So the path, the way she asks it: testis, epididymis, vas deferens, ampulla, ejaculatory duct, then the prostatic urethra, and out through the penis.

The traps. Sperm mature in the epididymis, not the testis. The dartos is smooth muscle and the cremaster is skeletal: her quiz offers both the wrong way round. And the cremaster lifts the testes in the cold, not in the heat.

Once more. The male system, structure by structure.

The scrotum keeps the testes three degrees cooler, with the dartos wrinkling the skin and the cremaster lifting the testes. The testes make sperm in the seminiferous tubules and testosterone in the Leydig cells, and the Sertoli cells nourish the sperm. The epididymis stores the sperm and matures them. The vas deferens carries them to the ampulla. The seminal vesicles add sixty per cent of the semen, alkaline, with fructose and prostaglandins, and form the ejaculatory ducts. The prostate, round the urethra under the bladder, adds a third, with citric acid and PSA. The bulbourethral glands clear acidic urine. The urethra carries urine and semen, and the penis, with its corpora cavernosa and corpus spongiosum, delivers the sperm.